MZ 510 — LP & IP formulation cheatsheet

red = new w.r.t. the base model (← base) · names in *italics* · ▸ = variant (lists only what it adds) · *cues* = wording that signals the model · §6 = recognition guide

Notation legend — one letter = one concept; superscripts separate same-letter items (f^P plant vs f^W WH fixed cost)

Vars: x quantity / flow; $y \in \{0,1\}$ decision (open, select); $z \in \{0,1\}$ helper (served by, AND); $n \in \mathbb{Z}_{\geq 0}$ number of (people, trucks); I inventory; u order (MTZ) / utilisation; v max-flow value; t start time; $e \geq 0$ excess / shortage / lateness
Params: k capacity; d demand; s supply; c unit (variable) cost, incl. transport; f fixed cost; h holding cost; p price; π unit profit / margin / score; a use per unit · b availability; r requirement; q fraction / share; τ time (transit, processing); δ due date · ρ release time; w weight · ℓ length; M big M (smallest valid bound)
Sets: uppercase set, lowercase index (P/p plants, W/w warehouses, J/j customers / products, T/t periods); never a set named I.

1 · General LP

1.1 Product mix / resource allocation

cues: limited resources (hours, material, budget, fuel), "how much of each to make", maximise profit / revenue, "at most X can be sold"

Sets • J (j): products / activities • R (r): resources
Params • π_j: unit profit (price − var. cost, or price × output per unit) • a_{rj}: use of r per unit of j • b_r: availability of r • d_j: max sales of j
Vars • $x_j \geq 0$: units of j (or time activity j runs)

$$\max \sum_j \pi_j x_j \quad \text{profit / revenue}$$
$$\text{s.t.} \sum_j a_{rj} x_j \leq b_r \quad \forall r \quad \text{resource capacity}$$
$$x_j \leq d_j \quad \forall j \quad \text{demand (market) cap}$$

▸ **Several plants, same % of capacity used**

Sets • P (p): plants • J (j): products
Params • k_p: capacity of plant p (units) • k_p^A: second plant limit, e.g. storage area • a_j: area per unit of j • π_j: unit profit; d_j: max sales
Vars • $x_{pj} \geq 0$: units of j at plant p • $u \in [0,1]$: common utilisation share (alt.)

$$\max \sum_p \sum_j \pi_j x_{pj} \quad \text{profit}$$
$$\text{s.t.} \sum_j x_{pj} \leq k_p \quad \forall p \quad \text{plant capacity}$$
$$\sum_j a_j x_{pj} \leq k_p^A \quad \forall p \quad \text{storage space}$$
$$\sum_p x_{pj} \leq d_j \quad \forall j \quad \text{demand}$$
$$\tfrac{1}{k_p}\sum_j x_{pj} = \tfrac{1}{k_1}\sum_j x_{1j} \forall p \neq 1 \quad \text{same \% of capacity}$$
$$\text{or } \sum_j x_{pj} = k_p u \quad \forall p \quad \text{same share } u \text{ (helper)}$$

▸ **Assemble a product from parts: max complete units**

Sets • J (j): parts • L (ℓ): departments / machines
Params • r_j: parts j needed per product • $a_{j\ell}$: hours per part j in ℓ • b_ℓ: hours available in ℓ
Vars • $x_{j\ell} \geq 0$: parts j made in ℓ • $x^F \geq 0$: finished products

$$\max \; x^F \quad \text{complete units}$$
$$\text{s.t.} \sum_\ell x_{j\ell} \geq r_j x^F \quad \forall j \quad \text{parts requirement}$$
$$\sum_j a_{j\ell} x_{j\ell} \leq b_\ell \quad \forall \ell \quad \text{department hours}$$

1.2 Allocation with average / ratio / share limits

cues: split a budget, "average risk / quality ≤", "at least X % in", "no more than the amount in", money left over earns a fixed rate

Sets • J (j): investment / allocation categories
Params • π_j: return rate of j • π^0: return of the leftover option • σ_j: risk (or quality) score of j • $\overline{\sigma}$: max average score • b: budget • q: min share of category k
Vars • $x_j \geq 0$: amount put in j • $x^0 \geq 0$: amount left in the leftover option

$$\max \sum_j \pi_j x_j + \pi^0 x^0 \quad \text{total return}$$
$$\text{s.t.} \sum_j x_j + x^0 = b \quad \text{budget}$$
$$\sum_j (\sigma_j - \overline{\sigma}) x_j \leq 0 \quad \text{average score (linearised ratio)}$$
$$x_k \geq q \sum_j x_j \quad \text{min share of k}$$
$$x_2 + x_3 \leq x_1 \quad \text{relative limit (e.g. A + B ≤ C)}$$

Ratio trick (a_j: score of j, c: limit): $\left(\sum_j a_j x_j\right) / \left(\sum_j x_j\right) \leq c$? $\sum_j (a_j - c) x_j \leq 0$ (valid because the denominator is ≥ 0). Check whether the average is over **all** money or only the invested part. Average return per $ = objective / b.

1.3 Blending

cues: mix inputs into products, "contains at least / at most X % of", inputs bought at a price, products sold at a price

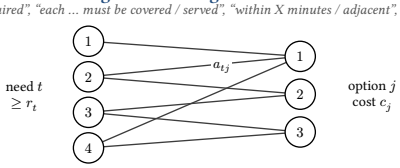
Sets • R (r): raw inputs • J (j): products / blends
Params • c_r: purchase cost of r • s_r: max supply of r • p_j: sale price of j • d_j: max demand of j • q_{rj}: min fraction of r in j • $q_{rj}^{\max}$: max fraction of r in j
Vars • $x_{rj} \geq 0$: amount of r used in j

$$\max \sum_j p_j \sum_r x_{rj} - \sum_r c_r \sum_j x_{rj} \quad \text{sales revenue − purchase cost}$$
$$\text{s.t.} \sum_j x_{rj} \leq s_r \quad \forall r \quad \text{supply}$$
$$\sum_r x_{rj} \leq d_j \quad \forall j \quad \text{demand}$$
$$x_{rj} \geq q_{rj} \sum_{r'} x_{r'j} \quad \forall r,j \quad \text{min fraction (quality)}$$
$$x_{rj} \leq q_{rj}^{\max} \sum_{r'} x_{r'j} \quad \forall r,j \quad \text{max fraction}$$

2 inputs: $x_{1j} \geq q\left(x_{1j} + x_{2j}\right)$? $(1-q)x_{1j} - q x_{2j} \geq 0$. Helper totals $x_j^T = \sum_r x_{rj}$ keep rows short (§4). Extra input bought at a stepped price ⇒ 3.6.

1.4 Covering: diet · shift scheduling · set covering

cues: "at least / minimum required", "each … must be covered / served", "within X minutes / adjacent", minimise cost or number used



Sets • T (t): needs (nutrients / periods / areas) • J (j): options (foods / shifts / sites)
Params • c_j: unit cost of option j • a_{tj}: amount of need t covered by one unit of j • r_t: requirement of need t
Vars • x_j: amount of option j — only the **domain** changes between variants

$$\min \sum_j c_j x_j \quad \text{cost (or count, } c_j = 1\text{)}$$
$$\text{s.t.} \sum_j a_{tj} x_j \geq r_t \quad \forall t \quad \text{requirement covered}$$

▸ **Diet / nutrition (continuous)**

$x_j \geq 0$: amount of food j; a_{tj}: nutrient t per unit of j; r_t: minimum intake.

▸ **Staffing with overlapping shifts**

Vars • $n_j \in \mathbb{Z}_{\geq 0}$: people **starting** shift j (replaces x_j)

$a_{tj} = 1$ if shift j works in period t. Shift = 2 consecutive periods ⇒ $n_{t-1} + n_t \geq r_t$ round-the-clock ⇒ cyclic ($n_0 = n_{|T|}$).

▸ **Workers do a combination of shifts, priced per combination**

Sets • Q: allowed pairs (j, j'), $j < j'$
Params • $c_{jj'}$: daily pay of pair, + bonus if not consecutive $(j' > j+1)$
Vars • $n_{jj'} \in \mathbb{Z}_{\geq 0}$: workers doing shifts j and j'

$$\min \sum_{(j,j') \in Q} c_{jj'} n_{jj'} \quad \text{labour cost}$$
$$\text{s.t.} \sum_{(j,j') \in Q : t \in \{j,j'\}} n_{jj'} \geq r_t \quad \forall t \quad \text{workers on duty in period t}$$

▸ **Locate sites so every area is reached within a time / distance**

Sets • J (j): candidate sites • T (t): areas to protect / serve
Params • τ_{jt}: travel time $j \to t$ (may be asymmetric) • $\overline{\tau}$: max allowed time • pop_j: eligibility data (e.g. population); $\overline{\text{pop}}$: threshold • c_j: site cost (1 if equal)
Vars • $y_j \in \{0,1\}$: open site j • $z_{jt} \in \{0,1\}$: site j serves t

$$\min \sum_j c_j y_j \quad \text{number / cost of sites}$$
$$\text{s.t.} \sum_j z_{jt} \geq 1 \quad \forall t \quad \text{every area served}$$
$$z_{jt} \leq y_j \quad \forall j,t \quad \text{only from an open site}$$
$$\tau_{jt} z_{jt} \leq \overline{\tau} \quad \forall j,t \quad \text{within time (forces } z=0 \text{ if } \tau > \overline{\tau}\text{)}$$
$$(\text{pop}_j - \overline{\text{pop}}) y_j \geq 0 \forall j \quad \text{eligibility rule (forces } y_j = 0\text{)}$$

Short form (no z): $\sum_j a_{tj} y_j \geq 1$ with $a_{tj} = 1$ if j reaches t (in time, same / adjacent area).
Partitioning ("exactly once", e.g. pick routes so each location is on exactly one): $\sum_j a_{tj} y_j = 1$, often + $\sum_j y_j = b$ (number of vehicles). **Max covering** (limited sites): $\max \sum_t w_t z_t$, $z_t \leq \sum_j a_{tj} y_j$, $\sum_j y_j \leq b$ (w_t: weight of area t).

1.5 Cutting stock (patterns)

cues: cut standard lengths / rolls into pieces, "number of standard pieces used", "waste / trim loss"

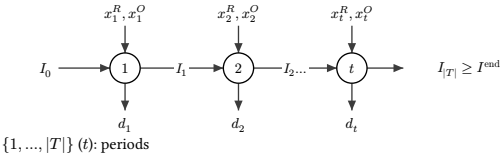
Sets • J (j): piece types • P (p): cutting patterns — list all p with $\sum_j \ell_j a_{jp} \leq k$
Params • k: standard length • ℓ_j: length of piece j • d_j: pieces of j ordered • a_{jp}: pieces j in pattern p • $c_p = k - \sum_j \ell_j a_{jp}$: trim waste of p
Vars • $n_p \in \mathbb{Z}_{\geq 0}$: standard pieces cut with pattern p • (b) $e_j \geq 0$: surplus pieces of j

$$\text{(a) } \min \sum_p n_p \quad \text{standard pieces used}$$
$$\text{s.t.} \sum_p a_{jp} n_p \geq d_j \quad \forall j \quad \text{demand}$$
$$\text{(b) } \min \sum_p c_p n_p + \sum_j \ell_j e_j \quad \text{trim waste + overproduction}$$
$$\text{s.t.} \sum_p a_{jp} n_p - e_j = d_j \forall j \quad \text{demand with surplus}$$

(b) = (a): total waste $= k \sum_p n_p - \sum_j \ell_j d_j$ is a constant plus the objective of (a), so both give the same optimum.

1.6 Multi-period production & inventory

cues: several periods / months / quarters, "carried over", "ending inventory of at least", regular vs overtime, holding cost



Sets • $T = \{1, \ldots, |T|\}$ (t): periods
Params • d_t: demand in t • k_t^R, k_t^O: regular / overtime capacity • c_t^R, c_t^O: unit costs • h_t: holding cost per unit • I_0: initial stock • I^{end}: required final stock • (b) c^B: backorder cost per unit
Vars • $x_t^R, x_t^O \geq 0$: regular / overtime production • $I_t \geq 0$: end inventory • (b) $I_t^+, I_t^- \geq 0$: stock / backlog

$$\min \sum_t \left(c_t^R x_t^R + c_t^O x_t^O\right) + \sum_t h_t I_t \quad \text{production + holding cost}$$
$$\text{s.t.} \; I_{t-1} + x_t^R + x_t^O - d_t = I_t \quad \forall t \quad \text{inventory balance}$$
$$x_t^R \leq k_t^R, \quad x_t^O \leq k_t^O \quad \forall t \quad \text{regular / overtime capacity}$$
$$I_{|T|} \geq I^{\text{end}} \quad \text{ending stock}$$

Costs rising by a fraction q per period: $c_t = c_1(1+q)^{t-1}$; holding as a fraction q' of production cost: $h_t = q' c_t$. "Demand met on time" ⇒ no backlog.

▸ **(b) Backlog allowed**

replace I_t by $I_t^+ - I_t^-$ in the balance; cost $h_t I_t^+ + c^B I_t^-$ *(holding + shortage)*; usually $I_{|T|}^- = 0$.

▸ **Several products sharing one capacity**

Sets • J (j): products
Params • $d_{jt}, c_{jt}, h_{jt}, I_{j0}, I_j^{\text{end}}$: per product • k_t: shared capacity
Vars • $x_{jt} \geq 0$: production • $I_{jt} \geq 0$: inventory

$$\min \sum_{j,t} \left(c_{jt} x_{jt} + h_{jt} I_{jt}\right) \quad \text{production + holding cost}$$
$$\text{s.t.} \; I_{j,t-1} + x_{jt} - d_{jt} = I_{jt} \quad \forall j,t \quad \text{balance per product}$$
$$\sum_j x_{jt} \leq k_t \quad \forall t \quad \text{shared capacity}$$
$$I_{j\,|T|} \geq I_j^{\text{end}} \quad \forall j \quad \text{ending stock}$$

▸ **Repeating cycle, consumption depends on the level (refuelling)**

Sets • T (t): stops / legs in a loop, $t = 0 \equiv |T|$
Params • c_t: price at stop t • ℓ_t: length of leg t • k: tank capacity • r: min reserve on arrival • g: extra burn per unit carried (from the burn formula)
Vars • $x_t \geq 0$: amount bought at t • $I_t^S, I_t^E \geq 0$: level at start / end of leg t

$$\min \sum_t c_t x_t \quad \text{purchase cost}$$
$$\text{s.t.} \; I_t^S = I_{t-1}^E + x_t \quad \forall t \quad \text{balance (loop closes, no start/end)}$$
$$I_t^S - I_t^E = \ell_t\left(1 + g\left(I_t^S + I_t^E\right)/2\right) \forall t \quad \text{burn } \propto \text{ average level: still linear}$$
$$I_t^S \leq k, \quad I_t^E \geq r \quad \forall t \quad \text{tank · reserve}$$

▸ **Production planning written as a transportation problem**

Sets • sources (t, m): period t, mode $m \in \{R, O\}$ • destinations: periods $t' \in T$
Params • k_t^m: capacity • c_t^m: unit cost • h: holding rate per period (% of cost) • $d_{t'}'$: net demand, $d_1' = d_1 - I_0$, $d_{|T|}' = d_{|T|} + I^{\text{end}}$
Vars • $x_{tmt'} \geq 0$: made in t with m for period t'

$$\min \sum_{t,m,t'} c_t^m (1 + h(t'-t)) x_{tmt'} \quad \text{production + holding cost}$$
$$\text{s.t.} \sum_{t'} x_{tmt'} \leq k_t^m \quad \forall t,m \quad \text{capacity (supply)}$$
$$\sum_{t,m} x_{tmt'} \geq d_{t'}' \quad \forall t' \quad \text{net demand}$$
$$x_{tmt'} = 0 \quad \forall t' < t \quad \text{no backorders (or } c = M\text{)}$$

Use when holding cost is a % of **the cost in the period the unit was made**. Capacity > net demand ⇒ dummy period, $c = 0$ (2.0).

1.7 Allocate time across price levels (markdowns) ← 1.1, time as the resource

cues: season of X weeks, demand depends on the price level, leftover sold at a salvage / outlet price

Sets • L (ℓ): price levels
Params • p_ℓ: price at level ℓ • d_ℓ: demand per week at p_ℓ • τ: season length (weeks) • s: stock available • p^0: salvage / outlet price
Vars • $x_\ell \geq 0$: weeks sold at level ℓ (integer if whole weeks)

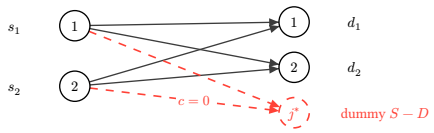
$$\max \sum_\ell p_\ell d_\ell x_\ell + p^0\left(s - \sum_\ell d_\ell x_\ell\right) \quad \text{in-season + salvage revenue}$$
$$\text{s.t.} \sum_\ell x_\ell \leq \tau \quad \text{season length}$$
$$\sum_\ell d_\ell x_\ell \leq s \quad \text{stock available}$$

The order of the weeks doesn't change revenue ⇒ schedule the levels from high to low (markdowns).

2 · Network models (totally unimodular ⇒ LP gives integer solutions)

2.0 Balanced vs unbalanced · dummy nodes (applies to all of §2)

cues: total supply ≠ total demand, "meet demand as much as possible", shortage penalty, a route that does not exist



Params • s_p: supply of source p • d_j: demand of destination j • $S=\sum_p s_p$: total supply • $D=\sum_j d_j$: total demand • c_j^e: shortage penalty per unit at j • M: big cost (forbidden arc)

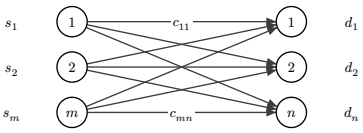
Vars • $x_{pj}\ge 0$: flow (2.1) • $x_{p,\text{ dum}}, x_{\text{dum},j}\ge 0$: flow to / from a dummy • $e_j\ge 0$: shortage at j (penalty form)

Balanced $S=D$	All rows **equalities**: $\sum_j x_{pj}=s_p$ *(supply)*, $\sum_p x_{pj}=d_j$ *(demand)*.
Supply > demand $S>D$	(a) supply rows ≤, demand rows = (or ≥); **or** (b) **dummy destination** with $d_{\text{dum}}=S-D$, cost $c_{p,\text{ dum}}=0$ (or idle cost) ⇒ all rows =. $x_{p,\text{ dum}}$ = **unused capacity**.
Demand > supply $S<D$	Plain model **infeasible**. (a) **dummy source** with $s_{\text{dum}}=D-S$, cost $c_{\text{dum},j}=c_j^e$ (0 if none) ⇒ $x_{\text{dum},j}$ = **unmet demand**; **or** (b) demand rows ≤, supply rows = ("meet demand as much as possible"); **or** (c) shortage variable: $\sum_p x_{pj}+e_j=d_j$, cost $c_j^e e_j$ ("penalty per unit short").
Forbidden pairs	$c_{pj}=M$ or $x_{pj}=0$ — incl. a destination that must be served in full (no dummy flow / no e_j for it).
Assignment (2.2)	More agents than tasks ⇒ dummy tasks ($c=0$) or agent rows ≤ 1. More tasks ⇒ dummy agents (task not done) or task rows ≤ 1.
Transshipment (2.3)	b_i = net supply of node i. $\sum_i b_i>0$ ⇒ dummy sink with $b_{\text{dum}}=-\sum_i b_i$, arcs from every source, $c=0$; or supply nodes ≤ b_i. $\sum_i b_i<0$ ⇒ dummy source with penalty.
Production as transp. (1.6)	Capacity > net demand ⇒ dummy period, $c=0$. Back-in-time arcs: $c=M$.

Exam tip: state the case (S vs D), then use inequalities **or** a dummy, not both. Report dummy flows as idle capacity / unmet demand.

2.1 Transportation

cues: sources with capacities ship to destinations with demands, cost per unit per route, "from … to …" cost table



Sets • P (p): sources (plants, refineries) • J (j): destinations

Params • s_p: supply of p • d_j: demand of j • c_{pj}: unit transport cost $p\to j$ (cost per mile × miles if given as distance)

Vars • $x_{pj}\ge 0$: units shipped $p\to j$

$$\begin{aligned} \min\ & \textstyle\sum_p\sum_j c_{pj}x_{pj} && && \text{transport cost}\\ \text{s.t.}\ & \textstyle\sum_j x_{pj}\le s_p && \forall p && \text{supply}\\ & \textstyle\sum_p x_{pj}\ge d_j && \forall j && \text{demand}\end{aligned}$$

Balanced / unbalanced, dummy nodes, shortages, forbidden arcs: see 2.0.

▸ Transportation with budget, minimum service %, deadlines

Sets • P: modes / sources • J: customer classes • $F\subset P\times J$: pairs too slow for the deadline

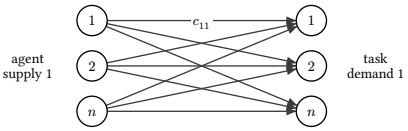
Params • s_p: capacity of mode p (vehicles × load; ∞ if unlimited) • d_j: demand of class j • q_j: min fraction of d_j served • b: budget

$$\begin{aligned} & \textstyle\sum_p x_{pj}\ge q_j d_j && \forall j && \text{service level}\\ & \textstyle\sum_{p,j} c_{pj}x_{pj}\le b && && \text{budget}\\ & x_{pj}=0 && \forall (p,j)\in F && \text{deadline (lead time too long)}\end{aligned}$$

"Is it feasible?" ⇒ check whether the minimum cost exceeds b (or drop the budget row and compare).

2.2 Assignment ← 2.1 with $s_p=d_j=1$

cues: "each … to exactly one", one-to-one matching, cost / time table agent × task



Sets • P (p): agents (people, machines) • J (j): tasks

Params • c_{pj}: cost / time if p does j

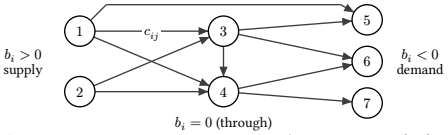
Vars • $y_{pj}\in\{0,1\}$: p does j (LP relaxation already integral)

$$\begin{aligned} \min\ & \textstyle\sum_p\sum_j c_{pj}y_{pj} && && \text{total cost / time}\\ \text{s.t.}\ & \textstyle\sum_j y_{pj}\le 1 && \forall p && \text{each agent at most once } (=1 \text{ if balanced})\\ & \textstyle\sum_p y_{pj}=1 && \forall j && \text{each task covered}\end{aligned}$$

More agents than tasks: agent rows ≤ 1, task rows = 1 (see 2.0).

2.3 Transshipment / min-cost flow ← 2.1 + intermediate nodes

cues: "through / via" warehouses, pumps, hubs; flows in and out of intermediate points; network drawing



Sets • N (i, j): nodes • $A\subset N\times N$: arcs (i,j)

Params • b_i: net supply (> 0 source, < 0 sink, 0 transshipment) • c_{ij}: unit cost on arc • k_{ij}: arc capacity

Vars • $x_{ij}\ge 0$: flow on arc (i,j)

$$\begin{aligned} \min\ & \textstyle\sum_{(i,j)\in A} c_{ij}x_{ij} && && \text{flow cost}\\ \text{s.t.}\ & \textstyle\sum_{j:(i,j)\in A} x_{ij}-\sum_{j:(j,i)\in A} x_{ji}=b_i && \forall i\in N && \text{flow conservation (out − in)}\\ & x_{ij}\le k_{ij} && \forall (i,j)\in A && \text{arc capacity}\end{aligned}$$

If $\sum_i b_i\neq 0$: see 2.0. Any arc is allowed (source → sink directly, or between intermediate nodes). "Where to add capacity / how much to invest" ⇒ shadow price of the supply row (§5).

▸ Path form, layered networks only — use arc form or path form

Sets • P (p): suppliers • W (w): transshipment nodes • J (j): customers

Params • s_p: supply • d_j: demand • c^P_{pw}, c^W_{wj}: unit costs per leg • k_w: node capacity; c_w: handling cost per unit

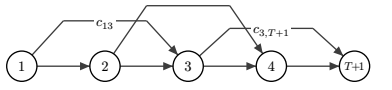
Vars • $x_{pwj}\ge 0$: units $p\to w\to j$

$$\begin{aligned} \min\ & \textstyle\sum_{p,w,j}\left(c^P_{pw}+c^W_{wj}+c_w\right)x_{pwj} && && \text{path cost}\\ \text{s.t.}\ & \textstyle\sum_{w,j} x_{pwj}\le s_p && \forall p && \text{supply}\\ & \textstyle\sum_{p,w} x_{pwj}\ge d_j && \forall j && \text{demand}\\ & \textstyle\sum_{p,j} x_{pwj}\le k_w && \forall w && \text{node capacity (if any)}\end{aligned}$$

No conservation rows: they are built into the path.

2.4 Shortest path · equipment replacement ← 2.3

cues: cheapest / shortest route from A to B, "keep or replace each year", trade-in value depends on age



Sets • N (i, j): nodes • A: arcs (i,j)

Params • $b_{\text{start}}=1$, $b_{\text{end}}=-1$, $b_i=0$ otherwise • c_{ij}: arc length / cost

Vars • $y_{ij}\in\{0,1\}$: arc used (relax to ≥ 0, TU)

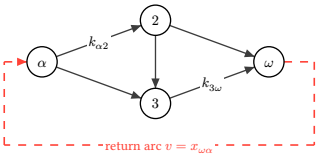
$$\begin{aligned} \min\ & \textstyle\sum_{(i,j)\in A} c_{ij}y_{ij} && && \text{path length}\\ \text{s.t.}\ & \textstyle\sum_{j:(i,j)\in A} y_{ij}-\sum_{j:(j,i)\in A} y_{ji}=b_i && \forall i\in N && \text{flow conservation } (+1\ /\ 0\ /\ -1)\end{aligned}$$

▸ Equipment replacement

nodes = years $1..T+1$; arc (i,j), $i<j$: buy at start of i, trade in at start of j; c_{ij} = price + maintenance for ages $0..j-i-1$ − trade-in at age $j-i$. Shortest path from 1 to $T+1$.

2.5 Maximum flow ← 2.3

cues: "maximum amount that can be sent / per hour", capacities on links, no costs



Sets • N (i, j): nodes, source α, sink ω • A: arcs

Params • k_{ij}: arc capacity

Vars • $x_{ij}\ge 0$: flow on arc • $v\ge 0$: total flow $\alpha\to\omega$

$$\begin{aligned} \max\ & v && && \text{throughput}\\ \text{s.t.}\ & \textstyle\sum_j x_{\alpha j}-\sum_j x_{j\alpha}-v=0 && && \text{source}\\ & \textstyle\sum_j x_{ij}-\sum_j x_{ji}=0 && \forall i\neq\alpha,\omega && \text{conservation}\\ & \textstyle\sum_j x_{\omega j}-\sum_j x_{j\omega}+v=0 && && \text{sink}\\ & x_{ij}\le k_{ij} && \forall (i,j) && \text{arc capacity}\end{aligned}$$

Same as: artificial arc $\omega\to\alpha$, max $x_{\omega\alpha}$, conservation everywhere; or max $\sum_j x_{\alpha j}$. **Two-way links** (lines, roads): arcs (i,j) and (j,i), each $\le k_{ij}$, or shared $x_{ij}+x_{ji}\le k_{ij}$.

3 · Integer programs

3.1 Fixed charge (setup / start-up / lease / entry cost) ← 1.1

cues: "if any … are produced, a fixed cost of", lease a machine, start-up cost, entry fee per market, "only if"

Sets • J (j): products / orders • R (r): resources

Params • π_j: unit margin (price − variable cost) • a_{rj}: use of r per unit of j • b_r: availability of r • d_j: max sales / order size • f_j: fixed cost if j is produced • M: big M = max possible x_j

Vars • $x_j\in\mathbb{Z}_{\ge 0}$ (or ≥ 0): units of j • $y_j\in\{0,1\}$: 1 if j is produced

$$\begin{aligned} \max\ & \textstyle\sum_j \pi_j x_j-\sum_j f_j y_j && && \text{margin − fixed cost}\\ \text{s.t.}\ & \textstyle\sum_j a_{rj}x_j\le b_r && \forall r && \text{resources}\\ & x_j\le My_j && \forall j && \text{produce only if set up}\end{aligned}$$

M = smallest valid bound, may differ per j: $\min_r b_r/a_{rj}$ (most you could make), or d_j (max sales / order). Capacity given as "% of capacity per unit": row $\sum_j a_j x_j\le 100\%$.

▸ Fixed charge + workforce sizing (full- / part-time, union rules)

Params • a_j^L: labour hours per unit • a_j^N, b^N: material per unit / available • c^{FT}, c^{PT}: wage per hour • τ^{FT}, τ^{PT}: hours per week per worker • $n^{\min}$: min full-time; q: full-time per part-time

Vars • x_j, y_j as above • n^{FT}, $n^{\text{PT}}\in\mathbb{Z}_{\ge 0}$: full- / part-time staff

$$\begin{aligned} \max\ & \textstyle\sum_j \pi_j x_j-\sum_j f_j y_j-\left(c^{\text{FT}}\tau^{\text{FT}}n^{\text{FT}}+c^{\text{PT}}\tau^{\text{PT}}n^{\text{PT}}\right) && && \text{profit − fixed − labour cost}\\ \text{s.t.}\ & \textstyle\sum_j a_j^N x_j\le b^N && && \text{material}\\ & \textstyle\sum_j a_j^L x_j\le \tau^{\text{FT}}n^{\text{FT}}+\tau^{\text{PT}}n^{\text{PT}} && && \text{labour hours}\\ & n^{\text{FT}}\ge n^{\min},\quad n^{\text{FT}}\ge qn^{\text{PT}} && && \text{union minimum · ratio}\\ & x_j\le d_j y_j && \forall j && \text{demand + setup } (M=d_j)\end{aligned}$$

Keep units consistent (all hours or all minutes).

▸ Fixed fee to enter a market + assortment rules

Sets • J (j): product lines / departments • G (g): candidate markets

Params • π_j: margin • f_g: entry cost of market g • k: total capacity • $M=k$ • b: max markets entered • q: max share

Vars • $x_{jg}\ge 0$: items of j sent to g • $y_g\in\{0,1\}$: enter market g

$$\begin{aligned} \max\ & \textstyle\sum_{j,g}\pi_j x_{jg}-\sum_g f_g y_g && && \text{margin − entry cost}\\ \text{s.t.}\ & \textstyle\sum_{j,g} x_{jg}\le k && && \text{capacity}\\ & \textstyle\sum_j x_{jg}\le My_g && \forall g && \text{ship only if entered}\\ & \textstyle\sum_g y_g\le b && && \text{at most b markets}\\ & x_{1g}=x_{2g}=x_{3g} && && \text{"equal numbers" in g}\\ & x_{jg}-q\textstyle\sum_{j'}x_{j'g}\le 0 && && \text{"at most q % of the total" (cross-multiplied)}\end{aligned}$$

3.2 Knapsack / capital budgeting ← 1.1, $x\in\{0,1\}$ or $\mathbb{Z}$

cues: "select which projects", yes/no investments, cash needed per year, NPV, indivisible items

Sets • J (j): projects / items • T (t): years / resources

Params • π_j: value (NPV, profit) of j • a_{tj}: cash used by j in year t • b_t: cash available in t

Vars • $y_j\in\{0,1\}$: select j (or $n_j\in\mathbb{Z}_{\ge 0}$: how many to buy)

$$\begin{aligned} \max\ & \textstyle\sum_j \pi_j y_j && && \text{total NPV / profit}\\ \text{s.t.}\ & \textstyle\sum_j a_{tj}y_j\le b_t && \forall t && \text{budget per year}\end{aligned}$$

Conditions between projects ("at most two", "only if", "not both") ⇒ §4. **LP relaxation** (drop integrality) gives a bound: ≥ IP optimum for max, ≤ for min. **Rounding the LP solution can be infeasible or far from optimal** ⇒ solve the IP (branch & bound).

3.3 Assign items to several roles with an "in both" count ← 3.2

cues: "exactly k must take part in both", choose members per event / role, score table

Sets • G (g): candidates • E (e): roles / events (here 2: A, B)

Params • π_{ge}: score of g in e • r: members per role • r^2: members in both roles

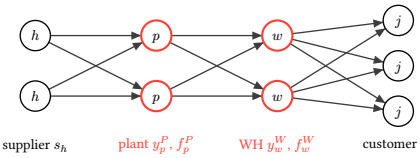
Vars • $y_{ge}\in\{0,1\}$: g takes role e • $z_g\in\{0,1\}$: g takes both

max	$\sum_g \sum_e \pi_{ge} y_{ge}$		total score
s.t.	$\sum_g y_{ge} = r$	$\forall e$	members per role
	$\sum_g z_g = r^2$		exactly r^2 in both
	$z_g \le y_{ge}$	$\forall g, e$	both ⇒ each
	$z_g \ge y_{gA} + y_{gB} - 1$	$\forall g$	each ⇒ both

Without the last row z_g could be 0 for someone in both roles ⇒ more than r^2 "doubles".

3.4 Facility location (plant & warehouse) ← 2.3

cues: open / close facilities, fixed annual operating cost, several levels (plants → warehouses → customers), "is it economical to keep all"



supplier s_h — plant y^P_p, f^P_p — WH y^W_w, f^W_w — customer d_j

Sets
- H (h): suppliers
- P (p): plants
- W (w): warehouses
- J (j): customers

Params
- s_h: supplier capacity
- k^P_p, k^W_w: plant / WH capacity
- d_j: demand
- $c^H_{hp}, c^P_{pw}, c^W_{wj}$: unit costs per leg
- f^P_p, f^W_w: fixed (annual) costs

Vars
- $x^H_{hp}, x^P_{pw}, x^W_{wj} \ge 0$: flows per leg
- $y^P_p, y^W_w \in \{0,1\}$: open plant / WH

min	$\sum_p f^P_p y^P_p + \sum_w f^W_w y^W_w +$		fixed cost
	$\sum_{h,p} c^H_{hp} x^H_{hp} + \sum_{p,w} c^P_{pw} x^P_{pw} + \sum_{w,j} c^W_{wj} x^W_{wj}$		transport (+ production)
s.t.	$\sum_p x^H_{hp} \le s_h$	$\forall h$	supplier capacity
	$\sum_w x^W_{wj} \ge d_j$	$\forall j$	demand
	$\sum_h x^H_{hp} - \sum_w x^P_{pw} \ge 0$	$\forall p$	plant balance (out ≤ in)
	$\sum_p x^P_{pw} - \sum_j x^W_{wj} \ge 0$	$\forall w$	WH balance
	$\sum_w x^P_{pw} \le k^P_p y^P_p$	$\forall p$	plant capacity (if open)
	$\sum_j x^W_{wj} \le k^W_w y^W_w$	$\forall w$	WH capacity (if open)

▸ Max number of open facilities + single sourcing of a customer

no suppliers; c^P_{pw} may include production.

Params
- $M = \sum_j d_j$: big M when a WH has no capacity
- $\hat{\jmath}$: customer that must be single-sourced
- b: max plants open

Vars
- $z_w \in \{0,1\}$: WH w serves customer $\hat{\jmath}$

	$\sum_w x^P_{pw} \le k^P_p y^P_p$	$\forall p$	plant capacity (if open)
	$\sum_p x^P_{pw} = \sum_j x^W_{wj}$	$\forall w$	WH conservation
	$\sum_j x^W_{wj} \le M y^W_w$	$\forall w$	ship only if WH open
	$\sum_w x^W_{wj} \ge d_j$	$\forall j$	demand
	$\sum_p y^P_p \le b$		at most b plants
	$x^W_{wj} \le M z_w \quad (M = d_{\hat{\jmath}})$	$\forall w$	customer $\hat{\jmath}$ only from chosen WH
	$\sum_w z_w = 1$		single sourcing (customer $\hat{\jmath}$)

Objective as above without the supplier term. Costs can be scaled (e.g. in k€) as long as all terms use the same unit.

▸ Keep or close existing facilities (one level)

facilities → markets only:

Params
- $f^W_w, k^W_w, c^W_{wj}, d_j$ as above

Vars
- $x^W_{wj} \ge 0$; $y^W_w \in \{0,1\}$

$\min \sum_w f^W_w y^W_w + \sum_{w,j} c^W_{wj} x^W_{wj}$ s.t. *demand* $\sum_w x^W_{wj} \ge d_j$, *capacity* $\sum_j x^W_{wj} \le k^W_w y^W_w$ (no capacity given ⇒ M). "Is it worth keeping all?" ⇒ solve with all $y^W_w = 1$ (plain transportation) vs y^W_w free and compare. At least $\left\lceil \sum_j d_j / k^W \right\rceil$ must stay open (equal capacities k^W). Opening rules ("if A then B", "at most 2", "A or B") ⇒ §4.

▸ Path variables + single sourcing + exactly b open

Params
- k^P_p: plant capacity
- d_j: demand
- c_{pwj}: unit cost $p \to w \to j$
- c^W_w: handling cost per unit at w
- f^W_w: fixed cost of w
- b: number of facilities to open

Vars
- $x_{pwj} \ge 0$: units $p \to w \to j$
- $y^W_w \in \{0,1\}$: open w
- $z_{wj} \in \{0,1\}$: w serves j

min	$\sum_{p,w,j} (c_{pwj} + c^W_w) x_{pwj} + \sum_w f^W_w y^W_w$		transport + handling + fixed
s.t.	$\sum_{w,j} x_{pwj} \le k^P_p$	$\forall p$	plant capacity
	$\sum_p x_{pwj} = d_j z_{wj}$	$\forall w, j$	demand from the chosen facility
	$\sum_w z_{wj} = 1$	$\forall j$	single sourcing
	$z_{wj} \le y^W_w$	$\forall w, j$	only an open facility
	$\sum_w y^W_w = b$		exactly b open

3.5 Capacity bought in steps from a table + entry fees ← 1.1

cues: table "investment → resulting capacity", "in multiples of", fee per product per market, "send at least / equally split"

Sets
- J (j): products
- G (g): markets
- L (ℓ): rows of the investment table

Params
- π_j: unit margin
- k_ℓ: total capacity at level ℓ
- f_ℓ: investment cost of level ℓ
- f^E: entry fee per product per market; b^E: fee budget
- r: min total volume
- d_{jg}: demand cap
- r_g: min per product in market g
- M: big M (largest capacity)

Vars
- $x_{jg} \ge 0$: units of j to g
- $y_\ell \in \{0,1\}$: choose level ℓ
- $y_{jg} \in \{0,1\}$: j introduced in g

Alternative A — one binary per level (exact for any capacity table)

max	$\sum_{j,g} \pi_j x_{jg} - \sum_\ell f_\ell y_\ell - f^E \sum_{j,g} y_{jg}$		margin − investment − entry fees
s.t.	$\sum_\ell y_\ell = 1$		choose one level
	$\sum_{j,g} x_{jg} \le \sum_\ell k_\ell y_\ell$		capacity
	$x_{jg} \le M y_{jg}$	$\forall j, g$	sell only if entered
	$f^E \sum_{j,g} y_{jg} \le b^E$		fee budget
	$\sum_{j,g} x_{jg} \ge r$		min volume
	$x_{jg} \le d_{jg}; \quad x_{j1} = x_{j2}$ (equal split)		demand; split rules
	$x_{jg} \ge r_g$	$\forall j$	min per product in g

Alternative B — one integer per block; replaces y_ℓ, use A or B

Sets
- S (s): blocks where each step adds the same capacity

Params
- k_0: base capacity
- k_s: capacity gained per step in block s
- b_s: max steps in block s
- f^S: cost per step

Vars
- $n_s \in \mathbb{Z}_{\ge 0}$: steps taken in block s
- x_{jg}, y_{jg} as in A

max	$\sum_{j,g} \pi_j x_{jg} - f^S \sum_s n_s - f^E \sum_{j,g} y_{jg}$		margin − investment − fees
s.t.	$\sum_{j,g} x_{jg} \le k_0 + \sum_s k_s n_s$		capacity
	$n_s \le b_s$	$\forall s$	block width

Plus all rows of A except the y_ℓ ones. **B is valid only if** each extra step buys **less** capacity than the previous block ($k_1 > k_2 > \ldots$): the solver then fills the blocks in order by itself (same idea as convex segments, 3.6). If the gain per step rises or jumps ⇒ use A.

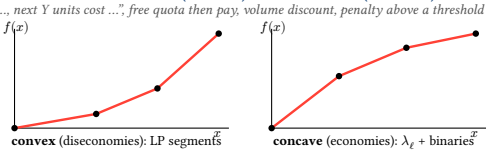
▸ Share a fixed number of units among items, value table per count

Vars
- $y_{\ell j} \in \{0,1\}$: item j gets ℓ units

$\max \sum_{\ell,j} \pi_{\ell j} y_{\ell j}$ ($\pi_{\ell j}$: value table) s.t. *one count per item* $\sum_\ell y_{\ell j} = 1$; *total units* $\sum_{\ell,j} \ell y_{\ell j} = b$. Needed when the value per extra unit is not constant.

3.6 Piecewise-linear cost: diseconomies (convex) vs economies (concave)

cues: "first X units cost …, next Y units cost …", free quota then pay, volume discount, penalty above a threshold



Which case? Unit cost **rises** with volume (convex, diseconomies) ⇒ simple segments, **no binaries**. Unit cost **falls** (concave, volume discount) ⇒ λ / y formulation **with binaries**.

▸ Convex cost (unit cost rises ⇒ no binaries)

Sets
- S (s): cost segments

Params
- k_s: width of segment s
- c_s: unit cost in s, $c_1 < c_2 < \ldots$

Vars
- $x_s \ge 0$: amount in segment s
- $x = \sum_s x_s$: total

$$\text{cost} = \sum_s c_s x_s \quad \text{with} \quad x = \sum_s x_s, \quad 0 \le x_s \le k_s \text{ the LP fills cheap segments first}$$

▸ Free quota, then pay per extra unit (water allowance, emission credits)

Sets
- J (j): activities
- R (r): other resources

Params
- π_j: revenue per unit
- a_{rj}, b_r: other resource use / availability
- a^Q_j: use of the quota resource per unit
- k_1: free quota; k_2: max extra (∞ if unlimited)
- c_2: price per extra unit

Vars
- $x_j \ge 0$: level of activity j
- $x_1, x_2 \ge 0$: free / paid use

max	$\sum_j \pi_j x_j - c_2 x_2$		revenue − extra cost
s.t.	$\sum_j a_{rj} x_j \le b_r$	$\forall r$	other resources
	$\sum_j a^Q_j x_j = x_1 + x_2$		quota resource balance
	$x_1 \le k_1, \quad x_2 \le k_2$		free part · extra part

Same as: x^T = total use of the quota resource, $x^T \ge \sum_j a^Q_j x_j$, $x^T \ge k_1$, cost $c_2(x^T - k_1)$. Area / mix shares on the activities ⇒ helper variable + ratio rows (§4).

▸ Convex cost inside a sourcing network

Sets
- P (p): plants
- J (j): demand points
- S (s): segments

Params
- d_j: demand
- c_{pj}: transport cost
- k_{ps}: width of segment s at plant p
- c_{ps}: unit cost in segment s (rising)

Vars
- $x_{pj} \ge 0$: shipped $p \to j$
- $x_{ps} \ge 0$: output of p in segment s

min	$\sum_{p,s} c_{ps} x_{ps} + \sum_{p,j} c_{pj} x_{pj}$		production (piecewise) + transport
s.t.	$\sum_j x_{pj} = \sum_s x_{ps}$	$\forall p$	output = shipments
	$x_{ps} \le k_{ps}$	$\forall p, s$	segment width
	$\sum_p x_{pj} \ge d_j$	$\forall j$	demand

▸ Concave cost / volume discount (binaries needed)

Sets
- $L = \{1..m\}$ (ℓ): breakpoints; segments $\ell = 1..m-1$

Params
- β_ℓ: breakpoint (quantity)
- $f(\beta_\ell)$: total cost at β_ℓ

Vars
- $\lambda_\ell \ge 0$: weight on breakpoint ℓ (slides: z)
- $y_\ell \in \{0,1\}$: x lies in segment ℓ

$f(x) = \sum_\ell \lambda_\ell f(\beta_\ell)$		cost replaced in objective
$x = \sum_\ell \lambda_\ell \beta_\ell$		x as a combination of breakpoints
$\sum_\ell \lambda_\ell = 1, \quad \sum_\ell y_\ell = 1$		convex comb.; one segment
$\lambda_1 \le y_1, \quad \lambda_\ell \le y_{\ell-1} + y_\ell, \quad \lambda_m \le y_{m-1}$		only the 2 ends of the chosen segment

Buying extra input at a stepped price inside a blend (1.3: x_{rj} input r in product j, s_r supply): x = amount of input r bought, supply row $\sum_j x_{rj} \le s_r + x$, objective $-\sum_\ell \lambda_\ell f(\beta_\ell)$, + the four rows above.

3.7 Single-machine scheduling with due dates

cues: jobs on one machine, processing times, due dates, "total delay / lateness", "cannot overlap"

Sets
- J (i, j): jobs

Params
- τ_j: processing time
- δ_j: due date
- M: big M ($\ge \sum_j \tau_j$)

Vars
- $t_j \ge 0$: start time of j
- $e_j \ge 0$: delay (lateness) of j
- $y_{ij} \in \{0,1\}$: 1 if i before j

min	$\sum_j e_j$		total delay (tardiness)
s.t.	$t_i + \tau_i \le t_j + M(1 - y_{ij})$	$\forall i < j$	either i before j …
	$t_j + \tau_j \le t_i + M y_{ij}$	$\forall i < j$	… or j before i (no overlap)
	$e_j \ge t_j + \tau_j - \delta_j$	$\forall j$	delay = max(0, completion − due)

▸ Positional alternative

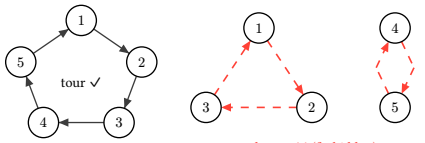
Vars
- $z_{j\ell} \in \{0,1\}$: job j in position ℓ
- $C_\ell \ge 0$: completion of position ℓ; $e_\ell \ge 0$ its delay

$$\sum_\ell z_{j\ell} = 1 \quad \forall j, \quad \sum_j z_{j\ell} = 1 \quad \forall \ell \qquad \text{assignment (2.2)}$$

$$C_\ell = \sum_{\ell' \le \ell} \sum_j \tau_j z_{j\ell'}, \quad e_\ell \ge C_\ell - \sum_j \delta_j z_{j\ell} \qquad \text{completion · delay}$$

3.8 Travelling salesman (TSP) ← 2.2

cues: visit every city once and return, shortest tour, "sequence of visits"



Sets
- N (i, j): nodes, node 1 = origin
- $Q \subset N$: any subset (subtour)

Params
- c_{ij}: distance $i \to j$

Vars
- $y_{ij} \in \{0,1\}$: j visited right after i
- u_i: visit order (MTZ)

min	$\sum_i \sum_j c_{ij} y_{ij}$		tour length
s.t.	$\sum_i y_{ij} = 1$	$\forall j$	enter once
	$\sum_j y_{ij} = 1$	$\forall i$	leave once
	$\sum_{i,j \in Q} y_{ij} \le \lvert Q \rvert - 1 \forall Q \subset N, \lvert Q \rvert > 1$		subtour elimination (DFJ)

▸ MTZ instead of DFJ

$u_1 = 1, \; 2 \le u_i \le \lvert N \rvert \; (i \ge 2), \; u_i - u_j + 1 \le (\lvert N \rvert - 1)(1 - y_{ij}) \; \forall i, j \ne 1.$

3.9 Consolidate orders on pre-defined routes vs. common carrier ← 2.2 + 3.1

cues: orders with release dates and deadlines, list of possible truck routes with fixed cost and capacity, LTL / common carrier alternative

Sets
- J (j): orders
- R (r): pre-defined routes
- $T = \{1..\lvert T \rvert\}$ (t): days

Params
- N_r: stops of route r
- dest$_j$: destination of j
- w_j: weight
- ρ_j: release day
- δ_j: deadline
- f_r: trip fixed cost
- k_r: truck capacity
- τ_{jr}: days to deliver j on r
- c_j: carrier cost of j; τ'_j carrier days

Vars
- $z_{jrt} \in \{0,1\}$: j on route r starting day t
- $y_{rt} \in \{0,1\}$: route r run on day t

- $z_j^L \in \{0,1\}$: j by carrier (only if $\rho_j + \tau_j' \le \delta_j$)

$$\begin{aligned}
\min\quad & \textstyle\sum_r \sum_t f_r y_{rt} + \sum_j c_j z_j^L && \text{trip fixed cost + carrier cost}\\
\text{s.t.}\quad & \textstyle\sum_{j:\rho_j \le t \le \delta_j - \tau_{jr},\, \text{dest}_j \in N_r} w_j z_{jrt} \le k_r y_{rt} \quad \forall r,t && \text{capacity, only if route runs}\\
& \textstyle\sum_{r:\text{dest}_j \in N_r} \sum_{t:\rho_j \le t \le \delta_j - \tau_{jr}} z_{jrt} + z_j^L = 1 \forall j && \text{each order shipped once}
\end{aligned}$$

Index ranges = **time windows** (released, on time) and **route covers the destination**. No carrier ⟹ drop z_j^L; days without trips (weekends) ⟹ $y_{rt} = 0$.

3.10 Batch shipping + per-unit option + transit inventory + one local source ← 2.3 + 3.1

cues: containers (full loads) vs per-unit shipping, lead time in days and an interest rate, rebalance between locations, choose one site for local production

Sets
- J (j, j'): locations (demand + candidate sites)
- L (ℓ): batch / container types

Params
- d_j: demand
- p^0: purchase price
- $c_{j\ell}$: freight per batch
- k_ℓ: units per batch
- c_j^L: per-unit freight
- τ_j: transit time (days)
- h: annual interest (holding) rate
- $c_{jj'}^R$: transfer cost per unit
- c_j^D: local unit cost
- k^D: fixed local volume

Vars
- $n_{j\ell} \in \mathbb{Z}_{\ge 0}$: batches
- $x_j^L \in \mathbb{Z}_{\ge 0}$: per-unit shipments
- $x_{jj'}^R \ge 0$: transfers
- $y_j \in \{0,1\}$: local site
- $x_j^D \ge 0$: local production

$$\begin{aligned}
\min\quad & \textstyle\sum_{j,\ell}(p^0 k_\ell + c_{j\ell}) n_{j\ell} + \sum_j (p^0 + c_j^L) x_j^L && \text{purchase + freight (batch, per unit)}\\
& \textstyle + \sum_j \frac{h}{365} p^0 \tau_j \left(\sum_\ell k_\ell n_{j\ell} + x_j^L\right) && \text{pipeline (in-transit) holding}\\
& \textstyle + \sum_{j,j'} c_{jj'}^R x_{jj'}^R + \sum_j c_j^D x_j^D && \text{transfers + production}\\
\text{s.t.}\quad & \textstyle\sum_\ell k_\ell n_{j\ell} + x_j^L + x_j^D + \sum_{j'} x_{j'j}^R \ge d_j + \sum_{j'} x_{jj'}^R \quad \forall j && \text{flow balance (in ≥ demand + out)}\\
& \textstyle\sum_j y_j = 1, \quad x_j^D = k^D y_j \quad \forall j && \text{one site, produces exactly } k^D
\end{aligned}$$

Pipeline cost = unit value × (annual rate / 365) × days in transit × units. Batches are integer; the per-unit option covers the rest. Build it up: direct shipping only ⟶ add transfers ⟶ add the local site.

4 · Logical-constraint toolbox

Symbols: $y_A, y_B, y_C, y_j \in \{0,1\}$ decisions (projects, sites, items A, B, C); $z \in \{0,1\}$ helper; $x \ge 0$ quantity; $g_1(x), g_2(x)$ linear expressions; b_1, b_2 right-hand sides; M big M = smallest valid upper bound.

Between binary decisions	
At least one	$\sum_j y_j \ge 1$ · two items ("either A or B"): $y_A + y_B \ge 1$
At most one (mutually exclusive)	$\sum_j y_j \le 1$ · two items: $y_A + y_B \le 1$
Exactly / at most / at least b of n	$\sum_j y_j = b \cdot \le b \cdot \ge b$ *(max plants, choose one level)*
If A then B (A only if B)	$y_A \le y_B$
If A then not B	$y_A + y_B \le 1$
Both or neither	$y_A = y_B$
If A or B then C	$y_A \le y_C$, $y_B \le y_C$ (or $y_A + y_B \le 2y_C$)
If A and B then C	$y_C \ge y_A + y_B - 1$
z = AND ("both")	$z \le y_A$, $z \le y_B$, $z \ge y_A + y_B - 1$ *(3.3)*
z = OR ("at least one")	$z \ge y_A$, $z \ge y_B$, $z \le y_A + y_B$
Linking continuous and binary	
If $x > 0$ then pay f / open	$x \le My$, cost fy *(fixed charge)*
Min lot r if produced	$ry \le x \le My$ *(semi-continuous)*
Either–or	$g_1(x) \le b_1 + My$, $g_2(x) \le b_2 + M(1-y)$ *(machine scheduling)*
If $g_1(x) > 0$ then $g_2(x) \ge 0$	$g_1(x) \le My$, $-g_2(x) \le M(1-y)$
At least b of m constraints hold	$g_{i(x)} \le b_i + M(1-z_i)$ for each constraint i, $\sum_i z_i \ge b$
Single sourcing	$x_{wj} \le M z_{wj}$ with $M = d_j$, $\sum_w z_{wj} = 1$
Integer from binaries	$x = \sum_\ell \ell y_\ell$, $\sum_\ell y_\ell = 1$ (discrete levels, step costs)
Choosing M (big M, all models)	smallest valid bound: $\min_r b_r/a_{rj}$, max order d_j, total demand $\sum_j d_j$, $\sum_j \tau_j$
Linearising	
Max / abs / excess	$e \ge g_1(x) - g_2(x)$, $e \ge 0$ (= $\max(0, \cdot)$) · $\lvert g_1(x)\rvert = e^+ + e^-$ with $g_1(x) = e^+ - e^-$
Helper (definition) variable	$x^T = \sum_j x_j$ (total area, total blend), then ratios use x^T: $x_j - 0.5x^T \le 0$, $x_j - 0.1x^T \ge 0$
Ratio / average / share	$\sum_j a_j x_j \le c \sum_j x_j \cdot x_k \ge q \sum_j x_j$ · equal %: $x_p^T/k_p = x_1^T/k_1$
Piecewise / non-linear cost	convex: segments, no binaries · concave: λ_ℓ weights + y_ℓ segment binaries (3.6)
Parameter test as a constraint	$(\text{pop}_j - \overline{\text{pop}})y_j \ge 0$ (forces $y_j = 0$ if $\text{pop}_j < \overline{\text{pop}}$) · $\tau_{jt} z_{jt} \le \overline{\tau}$ (forces $z_{jt} = 0$ if $\tau_{jt} > \overline{\tau}$) *(1.4)*
Forbid / force	$x_{pj} = 0$ (infeasible arc) · $y_j = 0$ if a rule fails

5 · Sensitivity report (Excel Solver)

Symbols: c_j objective coefficient of x_j; x_j^* optimal value; b_r right-hand side of constraint r; SP shadow price; Δ change; z^* optimal objective.

Final value	Variable: x_j^* · Constraint: resource **used** (LHS) at the optimum.
Reduced cost	Change in z^* per unit increase of a **zero-valued** variable = how much c_j must improve before x_j enters.
Allowable inc./dec. (objective c_j)	Range of c_j where the **optimal solution does not change**; $\Delta z^* = \Delta c_j \cdot x_j^*$. Outside: re-solve.
Shadow price (constraint)	Δz^* per unit increase of b_r: $\Delta z^* = \text{SP} \times \Delta b_r$. Value of one more unit of a scarce resource. Non-binding ⟹ SP = 0.
Allowable inc./dec. (RHS b_r)	Range where the **shadow price stays valid** — **not** where the solution stays the same: x^* values change.
100 % rule: objective	$\sum_j \Delta_j/\text{allowable}_j \le 1$ (allow. inc. for increases, dec. for decreases) ⟹ **optimal solution unchanged**; $\Delta z^* = \sum_j \Delta_j x_j^*$.
100 % rule: RHS	Same sum over changed $b_r \le 1$ ⟹ **shadow prices unchanged**; $\Delta z^* = \sum_r \text{SP}_r \Delta_r$.
Graphical (2 vars)	Plot constraints, feasible region, iso-profit line; optimum at a corner point.

Recipe. Sets ⟶ params (units) ⟶ vars (index, domain) ⟶ named objective terms ⟶ one named constraint family per sentence. %, ratios ⟶ cross-multiply; "if / only if / at most k", fixed fees ⟶ binaries (§4).

6 · Recognition guide: wording ⟶ model

If the problem says…	Use
full-time / part-time workers; union rules	3.1 ▸ workforce
fixed fee to enter a market / introduce a product	3.1 ▸ market entry · 3.5
select projects; yes/no; cash per year	3.2 knapsack
"exactly k in both" events / roles	3.3 · §4 AND
open / close plants or warehouses; fixed annual cost; several levels	3.4 location
"at most k facilities"; "only from one warehouse"	3.4 ▸ max open · single sourcing
"investment ⟶ resulting capacity" table; in multiples of	3.5 levels (A) or blocks (B)
"first X units cost …, next …"; free allowance then pay	3.6 convex (no binaries)
volume discount; price drops after X units	3.6 concave (binaries)
jobs, processing times, due dates; total delay	3.7 either-or with M
visit each city once and return	3.8 TSP
orders with release / deadline; candidate truck routes; carrier option	3.9 consolidation
containers vs per-unit; lead time + interest rate	3.10 batches + pipeline cost
"if A then B", "not both", "at least one of", "at most k of"	§4 binary logic
"either … or …", "if this then that" on amounts	§4 big M
"what if price / capacity changes", "how much would you pay for one more"	§5 sensitivity
limited hours / material / budget; "how much of each to make"; maximise profit	1.1 resource rows
"at most X can be sold", sales forecast	1.1 demand cap
"each plant uses the same %" of its capacity	1.1 ▸ same % (helper u)
"units of part A, B per product"; maximise number of complete products	1.1 ▸ assembly
"average … must not exceed"; "at least / no more than X % of the total"	1.2 · §4 ratio (cross-multiply)
"contains at least X %"; mix inputs into products	1.3 blending
"at least N needed in each period"; overlapping shifts	1.4 staffing
"each area must be within X minutes / adjacent"; fewest sites	1.4 set covering
"each … on exactly one route / in exactly one group"	1.4 partitioning
cut standard lengths into pieces; trim waste	1.5 cutting stock
months / quarters; "carried over"; ending inventory; overtime	1.6 balance rows
"backorders allowed", "shortages carried to next month"	1.6 ▸ backlog
a repeating loop (route, cycle); usage depends on the amount carried	1.6 ▸ cyclic
season of X weeks; demand depends on the price	1.7 price levels
total supply ≠ total demand; "as much as possible"; penalty per unit short	2.0 dummy / shortage
sources with capacities ⟶ destinations with demands; cost table	2.1 transportation
budget limit; "at least X % of class served"; deadline / lead time	2.1 ▸ service levels
"each person to exactly one task"	2.2 assignment
"through / via" hubs, warehouses, pumps	2.3 transshipment
shortest / cheapest route; "replace or keep each year"	2.4 shortest path
"maximum that can be sent", link capacities only	2.5 max flow
"if any are produced / opened, a fixed cost"; lease; start-up	3.1 fixed charge, $x \le My$